Past Paper Simulation (2025) — Paper 1

Paper 1 (Long Questions)

135 minutes (2.25 hours)

Total marks：105

FOR REVISION

For revision and self-assessment only. This is not an official examination paper.

Difficulty rating：**4.7** / 5.0

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INSTRUCTIONS

This paper consists of Section A(1), Section A(2) and Section B, each carrying 35 marks. Answer ALL questions and show all working. This AI-generated paper follows only the question structure of the 2025 paper with entirely new questions; it is not an official paper.

＊Independent, unofficial mock paper aligned to the DSE syllabus. Not affiliated with any official examination authority. For revision only.

Section A(1) (35 marks)

1. Simplify $\frac{(x^2y^{-3})^3}{x^{-5}y^2}$ and express your answer with positive indices.

(3 marks)

2. Simplify $\frac{5}{2x+3} - \frac{2}{x-1}$.

(3 marks)

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3. It is given that $a : b = 4 : 5$ and $3a + 2b = 110$. Find b.

(3 marks)

4. (a) The point $A(5, -2)$ is rotated clockwise about the origin through $90°$ to A'. Write down the coordinates of A'.

(b) A' is translated upwards by k units to A''. If A'', $B(1, 3)$ and $C(4, 9)$ are collinear, find k.

(3 marks)

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5. (a) Factorize $4ab - 12b$.

(b) Factorize $a^2 - 9$.

(c) Factorize $a^2 - 9 - 4ab + 12b$.

(4 marks)

6. (a) Solve the compound inequality $\frac{x+5}{2} < 1$ or $7 - 3x \geq 2x + 32$.

(b) Write down the greatest integer satisfying the compound inequality in (a).

(4 marks)

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7. (a) A shop sells a watch at a discount of **15%** on its marked price, and the selling price is **$1 020** . Find the marked price of the watch.

(b) The marked price of the watch is **60%** higher than its cost. Find the cost of the watch.

(c) If the shop sells the watch at a discount of **40%** on its marked price instead, will the shop make a profit? Explain your answer.

(5 marks)

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8. (a) In the figure, AD and BC intersect at E, where E is the mid-point of AD, and $AB \parallel CD$. Prove that $\triangle ABE \cong \triangle DCE$.

(b) G is a point on CD, and AG and BC intersect at H. It is given that $AB = 12$ cm, $DG = 8$ cm, $BC = 20$ cm and $AG = 16$ cm. Find the perimeter of $\triangle CGH$.

(5 marks)

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9. (a) The table shows the distribution of the numbers of game cards owned by the students in a class, where k is a positive integer.

Number of game cards:	0	1	2	3	4
Number of students:	3	5	k	9	4

It is given that the median of the distribution is **2** and the mode is **3** . Find the least possible value of k .

(b) Find the greatest possible value of k .

(c) Find the greatest possible standard deviation of the distribution.

(5 marks)

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Section A(2) (35 marks)

10. (a) Let $f(x) = 2x^3 + ax^2 + bx - 6$, where a and b are constants. It is given that $x + 3$ is a factor of $f(x)$, and the remainder when $f(x)$ is divided by $x - 2$ is 60. Find a and b.

(b) Someone claims that all the roots of the equation $f(x) = 0$ are rational. Do you agree? Explain your answer.

(6 marks)

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11. (a) It is given that $p(x)$ partly varies directly as x and partly varies directly as x^2. Suppose that $p(1) = -3$ and $p(3) = 3$. Find $p(x)$.

(b) If the equation $p(x) = c$ has two distinct real roots, where c is a constant, find the range of values of c.

(6 marks)

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12. (a) The stem-and-leaf diagram below shows the distribution of the weights (in kg) of **12** members of a fitness class before training, where k is an integer.

Stem (tens)	Leaf (units)
5	2 6 8
6	0 3 7
7	1 k k
8	4 6 9

It is given that the difference between the range and the inter-quartile range of the distribution is **16** kg. Find k.

(b(i)) After training, the distribution of the weights of the **12** members is represented by a box-and-whisker diagram, with least value, lower quartile, median, upper quartile and greatest value **50** kg, **57** kg, **66** kg, **74** kg and **85** kg respectively. Find the change in the upper quartile.

(b(ii)) Is the dispersion of the distribution of weights after training smaller than that before training? Explain your answer.

(7 marks)

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13. (a) The straight line L cuts the positive x-axis and the positive y-axis at A and B respectively. Let P be a moving point such that $PA = PB$. It is given that the equation of the locus Γ of P is $2x - y - 6 = 0$. Describe the geometric relationship between Γ and the line segment AB.

(b) Find the equation of L.

(c) Let Γ cut the x-axis at C, and M be the mid-point of AB. Find the area of the quadrilateral $OCMB$, where O is the origin.

(7 marks)

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14. (a) A solid right circular cone has base radius **12** cm and height **16** cm. It is cut by a plane parallel to its base at half of its height into a small cone and a frustum. Find the volume of the frustum in terms of π.

(b) Find the total surface area of the frustum in terms of π.

(c) The frustum is melted and recast into a solid cube. Assume that there is no loss of metal in the process. Is the total surface area of the cube less than that of the frustum? Explain your answer.

(9 marks)

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Section B (35 marks)

15. (a) A box contains **4** red balls, **3** blue balls and **5** green balls. If **3** balls are drawn randomly from the box at the same time, find the probability that the **3** balls are of the same colour.

(b) Find the probability that the **3** balls are all of different colours.

(4 marks)

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16. (a) Consider the simultaneous equations $\log_4(x^2) = 1 + \log_2 y$ and $(\log_2 x)(\log_2 y) = 6$, where $x > 0$ and $y > 0$. Let $u = \log_2 y$. Prove that $u^2 + u - 6 = 0$.

(b) Hence, find all the possible values of x.

(4 marks)

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17. (a) The common difference of an arithmetic sequence is $\mathbf{2}$, and its $\mathbf{1}$st term, $\mathbf{3}$rd term and $\mathbf{9}$th term are, in order, the first three terms of a geometric sequence. Find the first term of the arithmetic sequence.

(b) Find the least value of $\boldsymbol{n}$ such that the sum of the first $\boldsymbol{n}$ terms of the arithmetic sequence is greater than $\mathbf{1\,000\,000}$.

(6 marks)

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18. (a) Let $f(x) = x^2 - 2kx + k^2 + 3k$, where k is a positive constant. Using the method of completing the square, express the coordinates of the vertex V of the graph of $y = f(x)$ in terms of k.

(b(i)) The graph of $y = f(x)$ is translated leftwards by $2k$ units, and the vertex of the resulting graph is W. P is a point such that the orthocentre of $\triangle VWP$ is the origin O. Express the coordinates of P in terms of k.

(b(ii)) The coordinates of Q are $(0, 12)$. If V, W, P and Q are concyclic, find k.

(9 marks)

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19. (a) The equation of the circle C is $x^2 + y^2 - 4x - 2y + c = 0$, where c is a constant. It is given that the straight line $3x + 4y - 35 = 0$ touches C. By substituting $y = \frac{35 - 3x}{4}$ into the equation of C and using the fact that the resulting quadratic equation has equal roots, find c.

(b(i)) Let G be the centre of C. Two tangents are drawn from the point $P(11, 1)$ to C, touching C at A and B respectively. Find $\cos \angle APB$.

(b(ii)) Find AB in surd form.

(b(iii)) Someone claims that the radius of the circumcircle of $\triangle PAB$ is greater than 2 times the radius of its inscribed circle. Do you agree? Explain your answer.

(12 marks)

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